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LEARNING MODULE 01 | BLOCKCHAIN FUNDAMENTALS

What is Blockchain?

Core concepts, key properties and real-world applications.



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In this module: We explore what blockchain is, how it works, why it matters, and what it does – and does not – guarantee.

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From Records to Blocks

A blockchain organizes digital records into blocks and links them using cryptographic hashes, creating a sequence in which changes to earlier data can be detected.

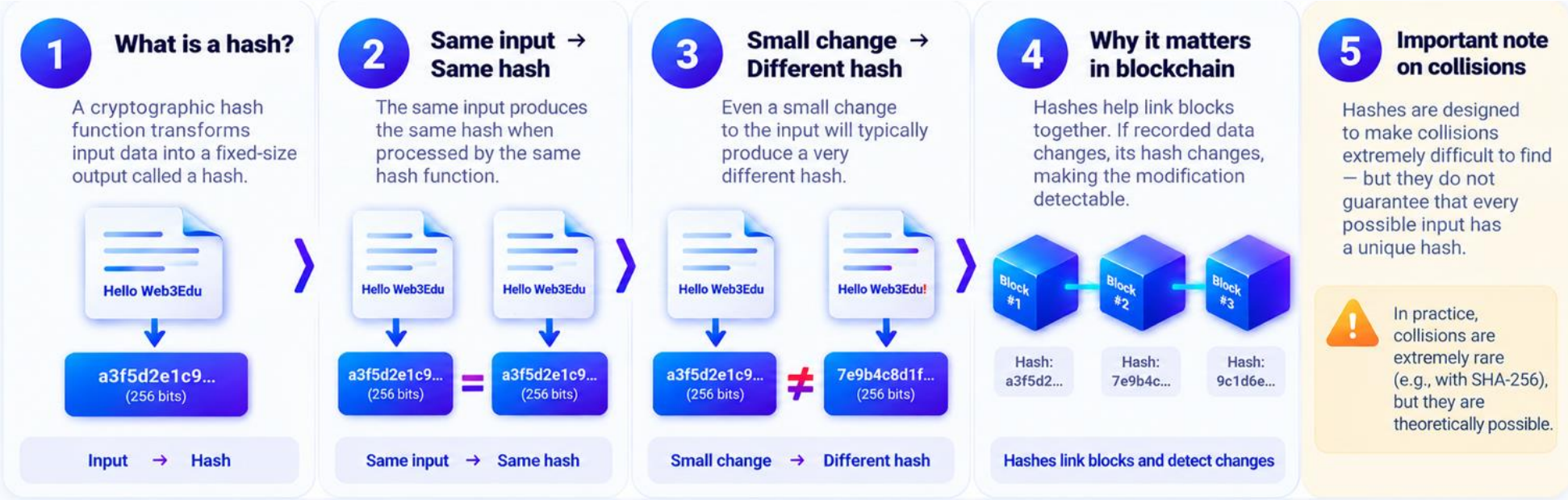


Key Takeaway: A blockchain organizes records into cryptographically linked blocks, making changes to the recorded history detectable.

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The Role of Hashes

A cryptographic hash function transforms input data into a fixed-size output called a hash.
Hashes help link blocks and make changes to previously recorded data detectable.



Key Takeaway: A hash acts like a compact cryptographic fingerprint of data, helping a blockchain detect changes without claiming that collisions are impossible.

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Linking the Blocks

Blocks are linked in sequence using cryptographic hashes. Each block contains its own data and a reference to the hash of the previous block.

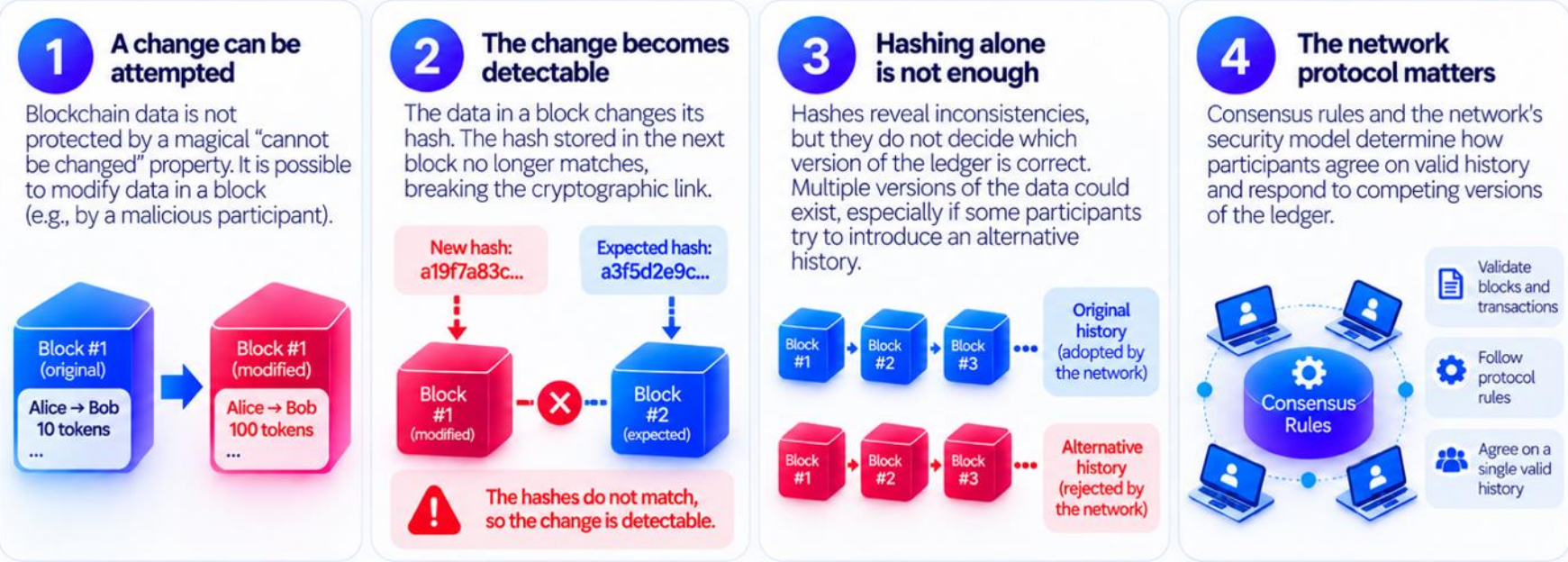


Key Takeaway: By storing references to previous block hashes, a blockchain creates a linked history in which changes to earlier data can be detected.

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Tamper-Evident Does Not Mean Immutable

Blockchain data can be changed, but such changes become detectable.
Whether the change is accepted depends on the network's protocol and security model.



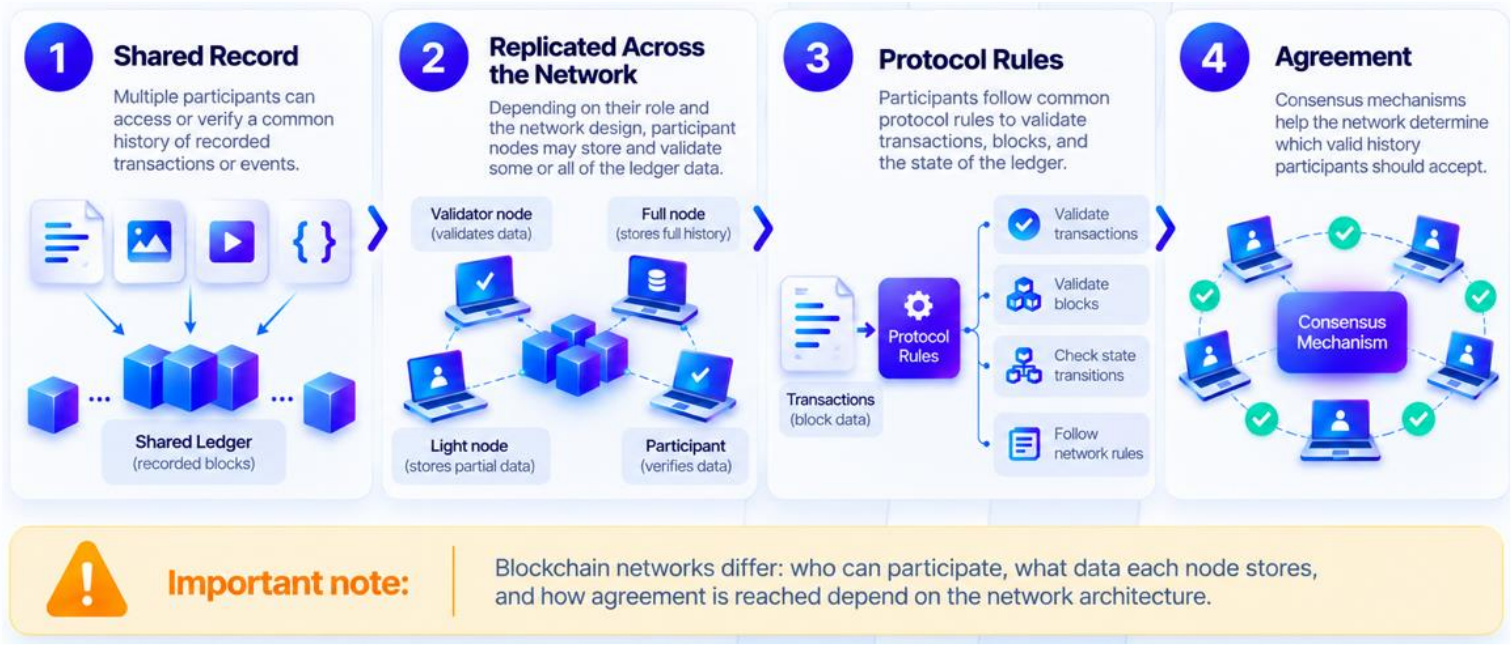
Key Takeaway:

Hash-linked blocks make tampering evident.
Resistance to rewriting blockchain history depends on more than hashing alone.

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One Ledger, Multiple Participants

Blockchain allows multiple participants to maintain and verify a shared record without relying on a single database as the only source of truth.



Important note: Blockchain networks differ: who can participate, what data each node stores, and how agreement is reached depend on the network architecture.

Key Takeaway: A blockchain can provide a shared, verifiable record across multiple participants — but its degree of decentralization depends on how the network is designed and governed.

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What Blockchain Gives Us

By combining cryptographic hashing, linked blocks, and a distributed network of participants, blockchain enables several important properties.

1 Tamper Evidence

Cryptographic links help make changes to previously recorded data detectable.



2 Traceability & Auditability

Recorded transactions and events can provide a history that authorized participants can inspect and verify, depending on the network design.



3 Shared State

Multiple participants can work with a common view of recorded data according to the network's protocol rules.



4 Reduced Reliance on a Single Authority

In appropriately designed networks, control and verification can be distributed across multiple participants rather than depend on one central operator.



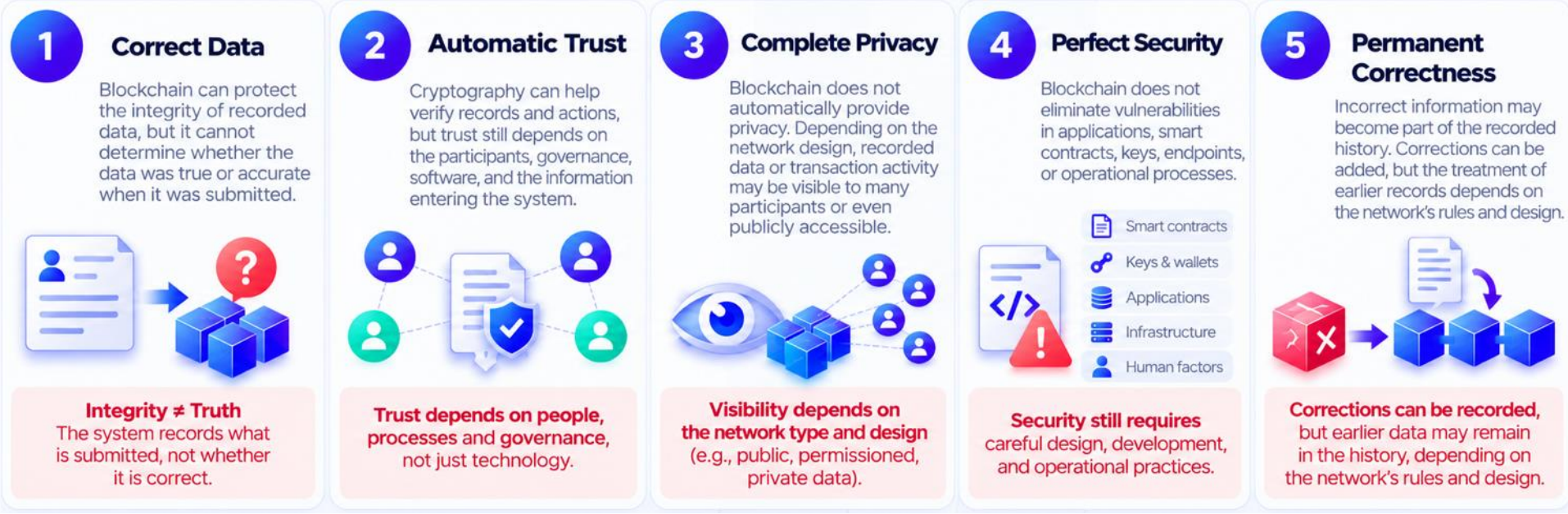
Important note: These are potential properties, not automatic guarantees. Their strength depends on the blockchain architecture, governance, and security model.

Key Takeaway: Blockchain can provide shared verification, traceability, and tamper evidence – when the network is designed to support those properties.

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What Blockchain Does NOT Guarantee

Blockchain can provide important properties, but it does not automatically solve all problems.
There are important limitations to understand.



Key Takeaway: Blockchain can strengthen integrity and verifiability, but it does not make data true, systems perfectly secure, or trust unnecessary.

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Blockchain Is NOT Cryptocurrency

Blockchain is a general technology for recording and verifying data. Cryptocurrency is one important application of that technology, but blockchain can support many other use cases.



Key Takeaway: Blockchain is the underlying technology; cryptocurrency is one category of applications built using blockchain and related distributed-ledger technologies.

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When Does Blockchain Make Sense?

Blockchain is useful when multiple parties need to share and maintain a trusted record without relying on a single central authority.

It is not the right solution for every problem. Understanding the right use cases helps us apply it effectively.

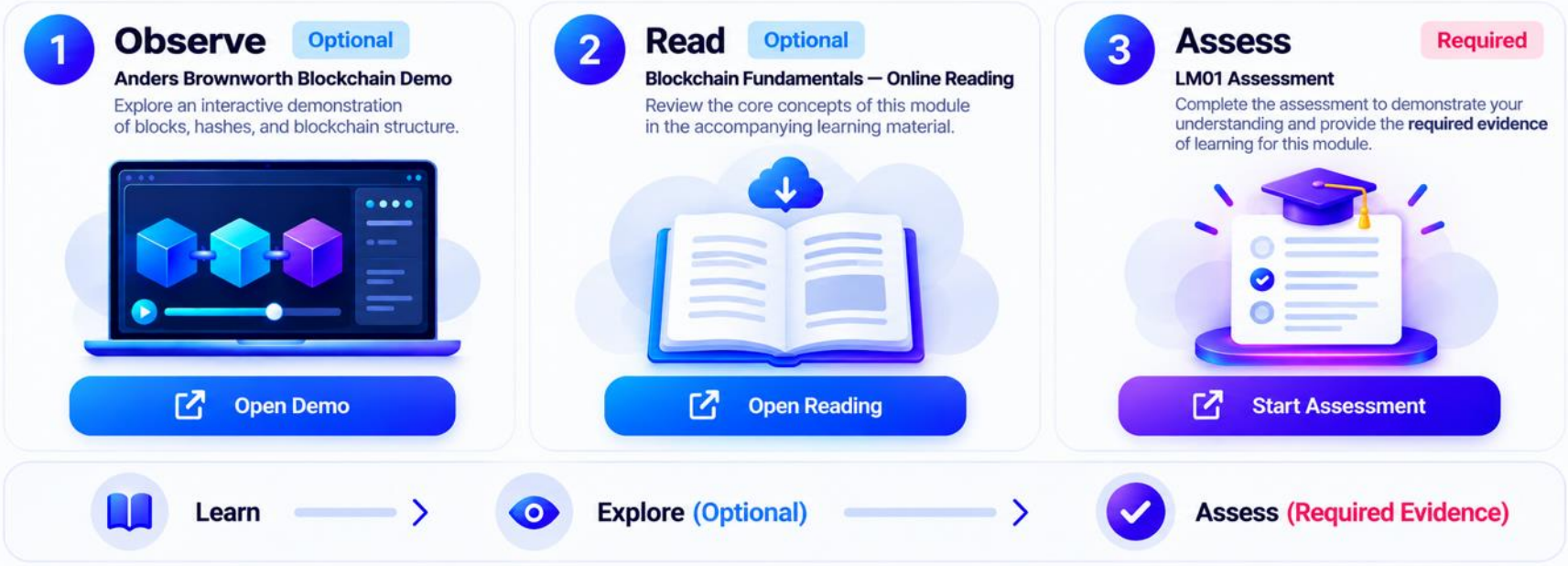


Key Takeaway: Use blockchain when you need a shared, tamper-evident and auditable record across multiple parties with limited mutual trust. Do not use it by default - simpler solutions may be more appropriate.

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From Concept to Exploration

You now understand the basic structure of a blockchain. The next step is to observe it, experiment with it, and see how changes affect the chain.



Key Takeaway: Understanding becomes stronger when you move from concepts to hands-on exploration.

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Key Takeaways

Blockchain is a powerful technology with unique properties, but it is not a solution for every problem.
Understanding both its capabilities and its limitations helps us use it correctly and set realistic expectations.

1 Shared, Tamper-Evident Record

Blockchain records data in blocks linked with cryptography, making unauthorised changes difficult.



2 Transparency and Auditability

Participants can verify the history of transactions, increasing trust and accountability (subject to the network design).



3 Not the Same as Cryptocurrency

Blockchain is the underlying technology. Cryptocurrency is one possible use case, not the same thing.



4 Benefits with Trade-offs

Blockchain can improve trust, reduce reliance on a single authority and enable new applications, but it also has trade-offs (e.g., performance, cost, complexity, privacy).



5 Use it When it Fits

Blockchain makes sense when multiple parties need a shared, trusted record. For many other problems, simpler solutions (e.g., traditional databases) are more appropriate.



6 Explore and Practice

The best way to build understanding is to experiment. Use the interactive demo and the Web3Edu simulator to see how blockchain works in practice.



Key Takeaway: Blockchain is a valuable tool when its strengths match the requirements of the problem.
Use it with a clear understanding of its limitations and always consider simpler alternatives.

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Visual Lesson Complete!

You now have the core concepts needed to explain what blockchain is, how blocks and hashes are connected, and what the technology can — and cannot — provide.



Next Steps

1. **Explore the interactive demo**
   Observe blocks, hashes and blockchain structure in action

2. **Experiment with the simulator**
   Build a chain, modify a block and observe what changes.

3. **Complete the assessment**
   Check your understanding of the key concepts from LM01.

Continue Your Learning Path
Continue with the LM01 activities in Web3Edu.

Key Takeaway: This visual lesson is your starting point. Continue exploring, experimenting and learning in Web3Edu.

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